

Integrable Systems — Exercise Sheet 9

Why the integrals commute

Variant: Questions only

Conventions, as in Lecture 9: $\{q, p\} = 1$; E_{ab} is the 2×2 matrix unit; for 2×2 matrices A, B , $A_1 = A \otimes \mathbb{1}$, $B_2 = \mathbb{1} \otimes B$, and $\{A_1, B_2\}$ is the 4×4 matrix whose entry in row (a, c) and column (b, d) is $\{A_{ab}, B_{cd}\}$, rows and columns ordered $(1, 1), (1, 2), (2, 1), (2, 2)$; so $(A \otimes B)_{(ac), (bd)} = A_{ab}B_{cd}$. P is the permutation, $P(v \otimes w) = w \otimes v$. The Toda site matrix is $L_n(\lambda) = \begin{pmatrix} \lambda - p_n & e^{-q_n} \\ -e^{q_n} & 0 \end{pmatrix}$, the monodromy matrix $T(\lambda) = L_N(\lambda) \cdots L_1(\lambda)$, and $t(\lambda) = \text{tr } T(\lambda)$.

Exercise 1 (the same bracket, computed twice). Take the oscillator's Lax matrix $L = \begin{pmatrix} p & q \\ q & -p \end{pmatrix}$ of Sheet 8, Exercise 1. (i) Compute $\{L_1, L_2\} = \sum_{a,b,c,d} \{L_{ab}, L_{cd}\} E_{ab} \otimes E_{cd}$ as a sum of terms $\pm E_{ab} \otimes E_{cd}$. (ii) Write the same object as a 4×4 matrix, and check its entry in row $(1, 2)$ and column $(2, 2)$ against the sum of (i). (iii) Write P as a 4×4 matrix. Check $(E_{ab} \otimes E_{cd})P = E_{ad} \otimes E_{cb}$ in two cases, and show that $P(A \otimes B)P = B \otimes A$.

Exercise 2 (the Toda site matrix, entry by entry). For one site, $L(\lambda) = \begin{pmatrix} \lambda - p & e^{-q} \\ -e^q & 0 \end{pmatrix}$. (i) Find the four non-zero brackets $\{L_{ab}(\lambda), L_{cd}(\mu)\}$. (ii) Assemble them into the 4×4 matrix $\{L_1(\lambda), L_2(\mu)\}$. (iii) Compute $L(\lambda) \otimes L(\mu) = L_1(\lambda)L_2(\mu)$ as a 4×4 matrix, then $[r(\lambda - \mu), L(\lambda) \otimes L(\mu)]$ with $r(u) = -P/u$, and check the quadratic Sklyanin bracket of Lecture 9 entry by entry.

Exercise 3 (the transfer matrix for two and three sites). (i) For $N = 2$ compute $t(\lambda) = \text{tr}(L_2(\lambda)L_1(\lambda)) = \lambda^2 + c_1\lambda + c_2$, and check that $H = \frac{1}{2}c_1^2 - c_2$ is the Hamiltonian $\frac{1}{2}(p_1^2 + p_2^2) + e^{q_1 - q_2} + e^{q_2 - q_1}$ of the two-particle ring. (ii) Lecture 8 found the $N = 2$ ring degenerate: its Lax matrix is $L = \begin{pmatrix} p_1 & a_1 + a_2 \\ a_1 + a_2 & p_2 \end{pmatrix}$ with $a_1 = e^{(q_1 - q_2)/2}$, $a_2 = e^{(q_2 - q_1)/2}$, and $\text{tr } L^2 = 2H + 4$. Show that nevertheless $\det(\lambda - L) = t(\lambda) - 2$. (iii) For $N = 3$ find $t(\lambda)$, and check $\det(\lambda - L) = t(\lambda) - 2$ for the ring's 3×3 Lax matrix of Lecture 8, $L = \begin{pmatrix} p_1 & a_1 & a_3 \\ a_1 & p_2 & a_2 \\ a_3 & a_2 & p_3 \end{pmatrix}$, $a_n = e^{(q_n - q_{n+1})/2}$, $q_4 = q_1$.

Where does the constant -2 come from?

Exercise 4 (involution for two sites, by hand). For the two-site ring, with $t(\lambda) = \lambda^2 + c_1\lambda + c_2$ as in Exercise 3: (i) Show that $\{t(\lambda), t(\mu)\} = (\lambda - \mu)\{c_1, c_2\}$. (ii) Compute $\{c_1, c_2\}$ directly, and conclude. (iii) What does the flow of c_1 do, and why does that make (ii) work?

Exercise 5 (the two-particle ring, separated). Carry out the separation Lecture 9 promises for $N = 2$: put $Q = \frac{1}{2}(q_1 + q_2)$, $P_Q = p_1 + p_2 = 2\bar{p}$, $x = q_1 - q_2$ and $p_x = \frac{1}{2}(p_1 - p_2)$. (i) Show that $\{Q, P_Q\} = \{x, p_x\} = 1$ and that the other four brackets among Q, x, P_Q, p_x vanish. (ii) Write H in the new variables. Show that the centre of mass moves uniformly, that (x, p_x) moves under $\mathcal{E} = p_x^2 + 2 \cosh x$, and that for $\mathcal{E} > 2$ the level set of $\mathcal{E}$ is a closed curve, so Lecture 6's action-angle variables apply to it. (iii) Show that $t(\lambda) = u^2 - \mathcal{E}$ with $u = \lambda - \bar{p}$. Find the bands $|t| \leq 2$ and the gap, and show that the gap closes exactly at the equilibrium.

Exercise 6 ($L(z)$ and the bands). (i) For $N = 3$, extend the eigenvalue recursion $a_{n-1}\psi_{n-1} + p_n\psi_n + a_n\psi_{n+1} = \lambda\psi_n$ periodically to all $n \in \mathbb{Z}$ and impose the Bloch condition $\psi_{n+3} = z\psi_n$, $z \neq 0$ (Lecture 9's reading). Show that it becomes $L(z)\psi = \lambda\psi$ for $\psi = (\psi_1, \psi_2, \psi_3)$, where $L(z)$ is L of Exercise 3 (iii) with $(L(z))_{13} = a_3/z$ and $(L(z))_{31} = a_3z$. Then show that $\det(\lambda - L(z)) = t(\lambda) - z - 1/z$. (ii) Show that $\text{tr} L(z)$ and $\text{tr} L(z)^2$ do not depend on z , and that $\text{tr} L(z)^3 = \text{tr} L^3 + 3(z + 1/z - 2)$. In what sense is this independent of z as a function on phase space? (iii) (*Computational.*) For $N = 6$ take the state $q = 0.35$ (0, 0.9, 0.5, -0.4, 0.3, -0.2), $p = 0.35$ (0.9, -0.5, 0.6, 0.2, -0.8, -0.4) with its mean subtracted. Compute $t(\lambda)$ by multiplying the site matrices, plot it with the lines $t = \pm 2$, and find the band edges as the eigenvalues of $L(1)$ and $L(-1)$. Integrate the motion to $t = 20$ and check that the band edges do not move while p_1 does.

Exercise 7 ★. *The open chain's linear bracket.* For the open two-particle Toda chain, $L = \begin{pmatrix} b_1 & a \\ a & b_2 \end{pmatrix}$ with $a = e^{(q_1 - q_2)/2}$ and $b_n = p_n$ (Lecture 8), Lecture 9's reading asks for a constant r with $\{L_1, L_2\} = [r, L_1 + L_2]$. (i) Compute the sixteen brackets $\{L_{ab}, L_{cd}\}$, show that $\{L_1, L_2\} = \frac{a}{2}[(E_{12} + E_{21}) \otimes (E_{11} - E_{22}) - (E_{11} - E_{22}) \otimes (E_{12} + E_{21})]$, and write it as a 4×4 matrix. (ii) Compute $[r, L_1 + L_2]$ for $r = \frac{1}{2}(E_{21} \otimes E_{12} - E_{12} \otimes E_{21})$ as a 4×4 matrix, and compare entry by entry. (iii) Prove that any L with $\{L_1, L_2\} = [r, L_1 + L_2]$, for a constant r , has $\{\text{tr} L^j, \text{tr} L^k\} = 0$ for all $j, k \geq 1$.